

# NIRUPAMA BALASUBRAMANIAN

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## EDUCATION

### University of Massachusetts, Amherst

Aug 2025 – May 2027

Master of Science, Computer Science, Specialization: Data Science, GPA: 4/4

Massachusetts, USA

- Relevant Coursework: Advanced NLP, Systems for Data Science, Advanced Machine Learning

### Vellore Institute of Technology

Sep 2021 – Aug 2025

Bachelor of Technology, Computer Science, GPA: 4/4

Chennai, India

## SKILLS

**Programming Languages:** Python, C, C++, Java, SQL, R, JavaScript, TypeScript

**Technology and Frameworks:** PyTorch, Pandas, Numpy, Scikit-learn, Spark, Docker, Redis, Flask, Streamlit

**Cloud:** AWS Certified: Cloud Practitioner, Solutions Architect- Associate

## PROFESSIONAL EXPERIENCE

### Travelers

Jun 2026 - Aug 2026

Data Science Intern - DSLDP

Hartford, USA

- Built a 3-tier pipeline to flag auto-insurance pricing disparities: A deterministic engine flags variables, a grounded LLM generates narratives, and an **agentic RAG chatbot** with tool calling, aids reasoning and report generation
- Delivered the solution as a Streamlit dashboard spanning across coverages and quarters; scaled to **20+ concurrent users** on a modular **plug-in architecture** that enables onboarding without disrupting production workflows
- Cut quarterly back-test review from **5 hours to 20 minutes** at **\$0.05** per query in LLM inference cost, with **100% recall** against analyst labels, enabling faster model refits and lower-latency pricing decisions for state teams

### Ascend Cargo Systems

Jan 2026 - May 2026

AI/ML Engineer Intern

San Francisco, USA

- Architected a production-grade agentic AI system with a custom Java Spring Boot **MCP server** and dual-LLM design, enabling real-time freight retrieval and autonomous carrier assignment across Slack, Microsoft Teams, email, and web chat, reducing manual assignment by **80%**
- Built an asynchronous document-processing system leveraging **GPT-4o Vision** & Microsoft **Graph API** to automate proof-of-delivery verification, achieving **10x** higher throughput & reducing latency from **3 hours to 10 min**

### Cogniverse Labs

Jan 2025 - Jul 2025

Data Science Intern

Bangalore, India

- Fine-tuned **Faster R-CNN** and **YOLOv8** models for India's Smart City roadside monitoring and deployed them using **TorchServe**. This computer vision (CV) project reduced manual inspection workload by **40%**

### National University of Singapore (NUS)

Dec 2023 – Jul 2024

Machine Learning Intern

Singapore, Singapore

- Developed a medical CV pipeline using **Vision Transformer** with progressive unfreezing and domain-specific augmentation achieving **96%** diagnostic accuracy; deployed real-time inference via a **Flask REST API**
- Architected and deployed an **SVM**-based churn prediction microservice for a Tier-2 telecom provider by engineering high-dimensional customer feature sets to achieve **92%** accuracy

## PROJECTS

### Multilingual Online Polarization Detection PyTorch, XLM-R, NLLB-200

- Engineered an XLM-RoBERTa framework for Polarization Detection across 9 languages using NLLB-200 for translation-based oversampling to resolve manifold collapse and increase **Macro F1** from **0.63** to **0.80**
- Integrated Supervised Contrastive Learning, Focal Loss and Dynamic Thresholding to disentangle overlapping rhetorical labels, achieving further **8%** boost in multi-label assignment

### Distributed ML Pipeline ML, Python, PySpark, Spark MLlib, Pandas

- Built a distributed flight-delay prediction pipeline on **1M+** records in **PySpark**, leveraging the Catalyst optimizer and lazy evaluation to scale **4x** faster than Pandas; tuned hyperparameters via grid search to achieve **0.88 AUC**

## PUBLICATIONS & ACHIEVEMENTS

 Diagnosis-based IQA and Enhancement for LDCT Images - Frontiers in Radiology, 2025 (PubMed Indexed)

 ANGEL: Automated Navigation Giving Emergency Location - IEEE I2ITCON 2025 (IEEE Xplore)

 Efficient Medical Diagnosis using ML: Cataract Detection - ICAECT 2024, Taylor & Francis (Scopus Indexed)

• **Winner:** Eco-Urban Nexus Hackathon 2024; **Finalist:** HackUMass 2025

• **NUS:** Winter 2023 - Valedictorian; Summer 2024 - AI Teaching Assistant for a cohort of 45 students